Lecture 24

3/16/26

ACTIVITY 8 Microscopy Review

Did Problems 1-5
Will do remaining problems next lecture

Ultrasound (US)

Began study of medical imaging by looking at ultrasound (US)

This is the only technique in course that does NOT use electromagnetic (EM) waves

US imaging uses high-frequency (1-15 MHz) sound waves
(Human hearing ≈ 20 kHz)

High-frequency sound is used because it gives better resolution (as w/ light microscopy)

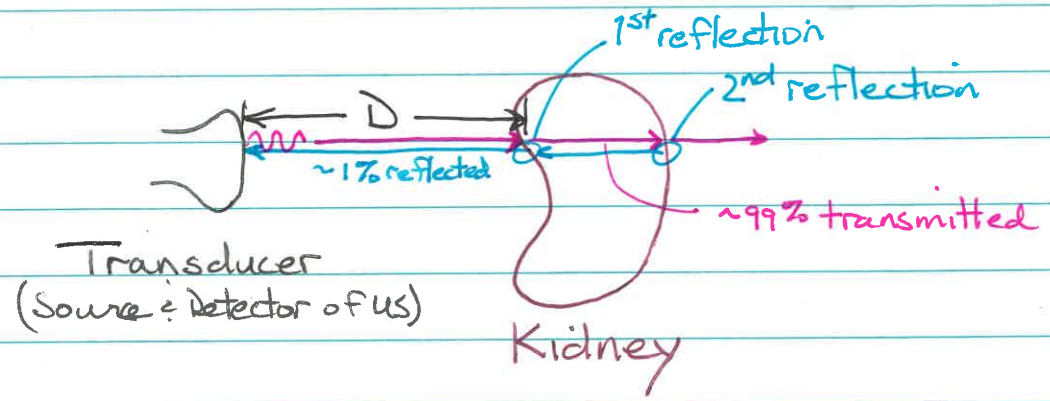
Higher frequencies ⇒ Shorter wavelengths

US resolution is on the order of tenths of mm (~100 cells) to 1 mm. This is typical for medical imaging techniques.

Medical imaging resolutions are larger than microscopy resolutions, but the body is larger than a cell, so the relative resolutions are analogous.

US imaging had its origins in the sinking of the Titanic & the subsequent development of SONAR.

Essence of Ultrasound



US imaging is "echo-based"

- Create pulse
- Detect echoes reflected off interfaces

Typically, there is ~1% reflection & ~99% transmission at each interface

We'll learn how to quantify amounts of reflection & transmission

Quantifying Depth

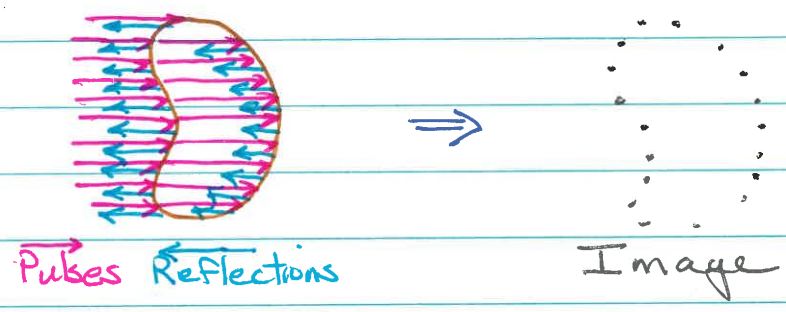
- Assume, in example, reflections only occur at the edges of the kidney
- Get reflections at different times due to different depths of two sides of kidney
- Measure time between creation of pulse & return of 1st & 2nd echoes
- Know time, calculate depth of tissue from kinematic formula:

$$\text{Distance} = \underbrace{\text{Speed}}_{\text{Known}} \times \underbrace{\text{Time}}_{\text{Measured}}$$

- Speed of sound in water = 1540 m/s
- The measured time is a "Go-Return time" that includes the time for the initial pulse to "go" to the interface & for the reflected pulse (echo) to "return" to the transducer
- The detected pulse has to travel twice the depth of the interface (1D going, 1D returning), so the kinematic equation can be rewritten by replacing "Distance" w/ "2D"

$$2D = (1540\ m/s)\cdot \text{time}^{echo} \Rightarrow \boxed{D = (1540\ m/s)\left(\frac{\text{time}^{echo}}{2}\right)}$$

- Send in a series of pulses to build up an image



This should give a feel for mapping depth

There are three issues of supreme importance in US imaging

(1) <u>Mapping tissue depth</u>

$$\boxed{D = \frac{1540\cdot \text{time}}{2}}$$

(2) <u>Image Contrast</u>

Echo intensity varies for different body parts

Q: What are the relative echo strengths of bone vs soft tissue?

A: Bone echoes are very strong
Soft tissue echoes are weaker

How is contrast encoded?

• Bright	Strong echoes	Bone
• Shades of gray	Intermediate echoes	Soft tissues
• Dark	No echoes	

Example] · Skull bright in fetal ultrasound
· Some US penetrates inside skull cavity & reflects off far side because fetal skull is relatively soft

(3) Image Resolution

US is a diffraction-limited technique

Resolution improves w/ increasing frequency
⇒ decreasing wavelength

US Strengths & Weaknesses

US Strengths

(1) Fast ⇒ capable of real-time imaging

(2) Very safe – uses sound (versus, e.g., ioniming radiation)

(3) Powerful method of monitoring blood flood & heart fuction (via Doppler US)

(4) Portable

(5) Relatively inexpensive

US Weaknesses

(1) Not good for bone or aerated tissue (e.g., lungs)

(2) Not good for deep imaging (especially w/ larger patients)

(3) Mostly anatomical imaging (vs functional)
Compare w/ nuclear imaging & MRI